Dr. D. Y. Patil Educational Federation's
Dr. D. Y. Patil College of Engineering & Innovation
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(An Autonomous Institute Affiliated to Savitribai Phule Pune University)
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Department of Computer Engineering

Program name :- Computer Engineering **Class:** S.Y. B. Tech **Div:-** C

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FinAI : AI-Powered Personal Finance & Wealth Advisor

Project Description :-

FinAI is a full-stack, AI-driven personal finance platform that provides automated budgeting, real-time financial health score diagnostics, income and expense tracking, milestone-based savings goals and an interactive financial advisor chat assistant. The platform helps salaried professionals, students, freelancers and household managers understand their spending, savings and financial goals without requiring advanced financial knowledge.

The system combines a React frontend with a Python Flask + SQLAlchemy backend (SQLite database) and an optional Node.js Express server. A built-in rule-based NLP advisor engine analyses income and expense data, calculates a 0-100 Financial Health Score, generates a personalised 50/30/20 budget plan and flags overspending categories. The AI Advisor chat lets users ask questions about loans, SIPs, emergency funds, tax brackets and budgets in natural language, and even log expenses through commands such as "Log $45 for groceries", with voice responses through Web Speech synthesis.

The application features a responsive neo-glassmorphism dark interface built around a Dashboard, Income & Expense Tracker, Budget Generator, Savings Goals, Analytics Charts, Monthly Report and the AI Advisor chat. Four pre-configured financial personas can be switched instantly, and interactive charts, animations, sound effects and confetti celebrations make everyday money management engaging.

Technology & AI :-

Frontend: React 19 / Vite / JavaScript (ES6+) / HTML5 / CSS3 / Tailwind CSS 3.4
Backend: Python Flask + SQLAlchemy, Node.js + Express 5

AI: Local rule-based NLP advisor engine and Financial Health Score algorithm
Database: SQLite (finai.db) via SQLAlchemy
Visualization & UI: Recharts, Framer Motion, Canvas Confetti, Lucide React
Browser APIs: Web Speech synthesis, Web Audio API, LocalStorage
Run: Vite dev server (5173), Flask (5000), Express (3001)

Key Features :-

- Automated 50/30/20 budget generator (Needs / Wants / Savings) with an interactive "What-If" simulator for projected annual savings
- Financial Health Score (0-100) based on savings rate, budget adherence, emergency runway and spending discipline
- AI Advisor chat covering loans, SIPs, emergency funds, tax brackets and budgets, with command-based expense logging and voice responses
- Income and expense ledger with categories, payment methods, filters and search
- Milestone-based savings goals with progress rings, monthly contribution targets and confetti on completion
- Interactive analytics: cashflow, category breakdown and budget vs actual charts (Recharts)
- Printable monthly executive financial report
- Four pre-configured personas (salaried professional, student, freelancer, household) with a scenario switcher
- Neo-glassmorphism dark UI with Framer Motion animations, Web Audio sound effects and LocalStorage state persistence

Scenarios :-

Scenario 1: Alex Morgan, a salaried professional with a fixed monthly salary of $6,500, logs income and daily expenses across categories such as dining, leisure and housing. The system analyses the spending data, identifies lifestyle inflation and overspending areas, and generates a personalised 50/30/20 budget while tracking progress towards a house down payment goal.

Scenario 2: Maya Chen, a college student, manages a tight monthly allowance of $850 across textbooks, the dining hall and discretionary spending. The platform helps her set realistic budget limits, track spending habits, run What-If cuts and build micro-savings towards tech hardware.

Scenario 3: Devon Vance, a freelancer with variable cashflow (about $7,400 a month), logs earnings from multiple clients and project expenses. The system sets aside a 25% tax safe-harbor vault, adapts budget recommendations to fluctuating income and builds a 6-month living runway buffer.

Scenario 4: The Sharma Family, a household with dual income ($9,800 a month), monitors shared income and expenses such as the mortgage, groceries and utilities. The platform raises grocery and utility overrun alerts, generates a consolidated monthly report and tracks a children's 529 education fund goal.

Milestones :-

Milestone 1 – Project Planning & Requirement Analysis

Define the objective of FinAI (FinWise), identify target users (salaried professionals, students, freelancers, household managers), finalize core features, and design the overall application workflow and architecture.

Milestone 2 – Backend & Data Model Setup

Build the Flask and SQLAlchemy backend with SQLite (finai.db), design the schema for User, Income, Expense, Budget and SavingsGoal, seed the four personas, and implement REST APIs; add the optional Node.js Express server.

Milestone 3 – Health Score & Budget Engine

Implement the Financial Health Score algorithm (savings rate, budget adherence, emergency runway, spending discipline) and the 50/30/20 budget generator with the What-If savings simulator.

Milestone 4 – AI Advisor & Frontend Build

Develop the rule-based NLP AI Advisor chat with voice responses and expense-logging commands, and build the React 19 + Vite + Tailwind frontend with Recharts analytics, Framer Motion animations and confetti effects.

Milestone 5 – Testing, Optimization & Documentation

Test the application across mobile and desktop devices, fix API and UI issues, improve performance and reliability, verify LocalStorage and database synchronisation, and document setup steps for the frontend (port 5173), Flask (5000) and Express (3001) servers.

Conclusion :-

FinAI (FinWise) provides a simple and interactive way to take control of personal finances. By combining income and expense tracking, an automated 50/30/20 budget generator, a Financial Health Score and an AI Advisor chat, the platform helps users understand their spending, savings and financial goals. The project demonstrates how modern full-stack web technologies (React, Flask, Node.js) and rule-based NLP can make personal finance management more accessible, disciplined and useful for everyday individuals.

Expected Outcome :-

FinAI aims to make personal finance management more accessible by reducing the manual complexity of budgeting and presenting income, expenses and savings in a simple, visual and understandable format. The platform combines automated budgeting, health score diagnostics, conversational financial guidance and printable monthly reports to create an engaging personal finance planning experience.